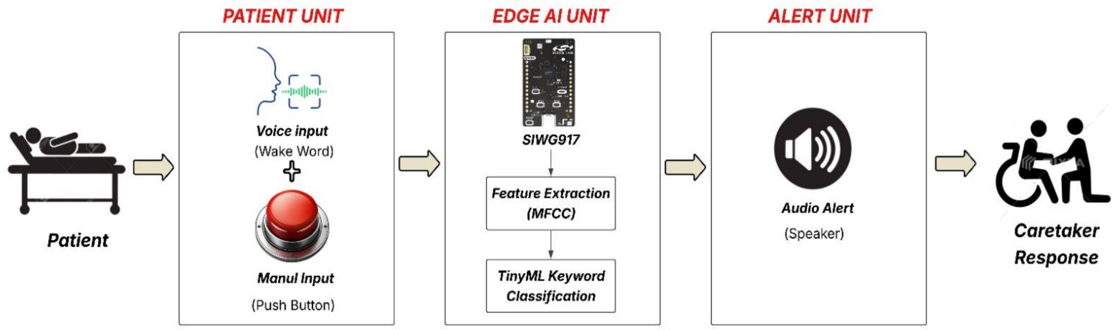
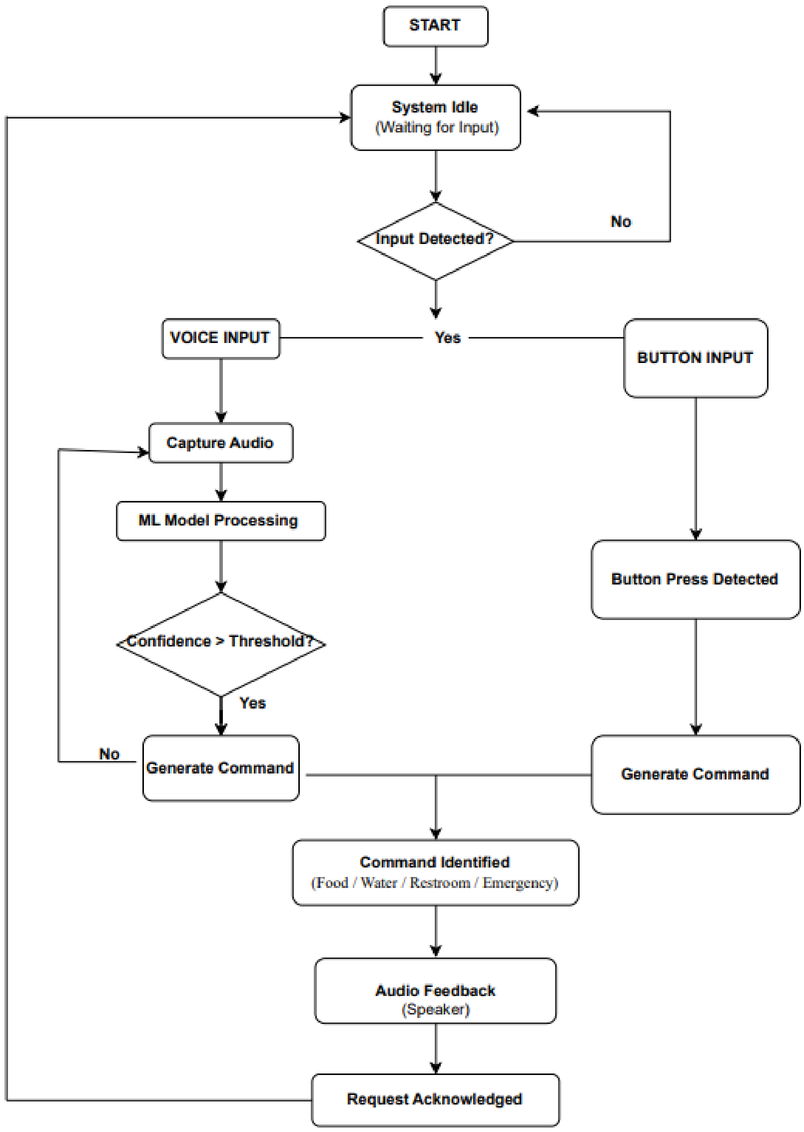
SYSTEM ARCHITECTURE & BLOCK DIAGRAM

1. HARDWARE BLOCK DIAGRAM



2. SOFTWARE ARCHITECTURE & DATA FLOW



3. INFERENCE LOCATION

- All audio processing including MFCC feature extraction and keyword classification runs entirely on the SiWG917 at the edge.
- The TensorFlow Lite Micro model is deployed and executed directly on-chip, with no dependency on external servers or cloud APIs for inference.
- The SiWG917 handles both inference and post-inference tasks, including audio feedback playback and result handling.
- The web server is responsible solely for dashboard rendering, request logging, and acknowledgment management it performs no machine learning computation.
- Inference latency is maintained under 200ms, achieved purely through efficient on-device processing.

4. DATA LEAVING THE DEVICE

- Raw audio is never transmitted it is captured, processed, and discarded entirely within the SiWG917.
- MFCC feature vectors are not transmitted they exist only as intermediate computations within the on-device inference pipeline.
- The only data transmitted is the interpreted command label (e.g., WATER, EMERGENCY) along with a timestamp forming a minimal and non-sensitive payload.
- No biometric, acoustic, or personally identifiable data leaves the device at any point.
- Caretaker acknowledgment and request status updates occur between the SiWG917, web server, and dashboard, and can be confined entirely within a local or institutional network when deployed on-premise.
- This architecture ensures privacy by design, as sensitive interaction data never reaches external servers, cloud platforms, or third-party systems.